Data Scientist Mission: Fix News Aggregator's Bizarre Recommendations

Use UMAP and DBSCAN to diagnose why financial news appears with entertainment gossip. Embedding quality diagnosis and cleanup plan for the 10,000-article corpus (all-MiniLM-L6-v2, 384-d).

Executive summary

Bottom line: invest in data infrastructure; do not abandon embedding-based personalization. The embedding model is behaving as designed. The damage traces to one identifiable slice of the corpus: **1,936 articles (19.4%) from two publishers, Generic News Co and Poor Quality Source**. All of them carry stub metadata (4-word descriptions such as "Short article about sports." and placeholder titles such as "Article 17") and 100 words or fewer of body text. The other 8,064 articles (nine reputable publishers) embed cleanly.

Key results

- **UMAP:** 8 compact, category-pure clusters with no cross-category overlap. Low-quality articles are interleaved inside every cluster rather than forming their own island.
- **DBSCAN on UMAP coordinates:** the prescribed defaults (eps=0.5, min_samples=5) flag **0** outliers, so the parameters had to be tuned. At eps=0.075, min_samples=10 the run flags **197 outliers (1.97%)** across 8 clusters, exceeding the 50+ target.
- **Source effect:** articles from the two low-quality publishers are outliers at **4.24%** versus **1.43%** for the rest (3.0x; chi-square $p<0.001$).
- **Embedding-space evidence is much stronger than the UMAP view:** low-quality articles sit **2.7x farther** from their category centroid (mean 1.04 vs 0.39). DBSCAN on the raw 384-d vectors isolates **1,922 articles with 100% precision and 99.3% recall** against the low-quality set.
- **Cleanup simulation:** removing the low-quality slice lifts category silhouette (cosine) from **0.66 to 0.84** and drops raw-space outliers from 1,922 to 0.

Honest caveats

- Source and article length are **perfectly confounded** in this extract (0 short articles outside the two sources; 0 low-quality-source articles above 100 words). The data cannot separate "bad publisher" from "too little text"; Section 4 proposes tests that can.
- Embedding neighbors never crossed categories here (top-10 neighbor category purity = 100%), and low-quality articles appear in only 0.10% of top-10 neighbor slots. So I **could not reproduce** "sports in tech feeds" directly from nearest-neighbor lists on this extract. The degradation shows up as diffuse, low-margin vectors for 19.4% of the catalog. How that turns into bad feeds (profile averaging, ANN approximation, ranking) is untested.
- The corpus is clearly synthetic (templated titles and repeated description text), so absolute numbers are cleaner than production would be. The method and reasoning transfer; the exact thresholds should be re-derived on live data.

1. Method and parameters

Data: `article_embeddings.npy` (10,000 x 384) and `article_metadata.csv` (article_id, title, description, category, source, word_count, publish_date). No nulls or empty descriptions. Note the metadata column is `source` (the brief calls it source publisher).

UMAP: n_neighbors=15, min_dist=0.1, n_components=2, random_state=42, as specified. DBSCAN: applied to the 2-D UMAP coordinates as instructed, then cross-checked in the original 384-d space.

Why DBSCAN needed tuning. UMAP packs each category into a tight blob about 1 unit wide, so the suggested eps=0.5 swallows everything into 7 clusters and returns zero noise points. I swept eps and min_samples and chose eps=0.075, min_samples=10, which yields a reviewable ~2% flag rate (197 articles) and recovers all 8 categories as separate clusters. This is a judgment call, not an optimum: the outlier count is very sensitive to eps (table below), so counts should be read as relative, not absolute.

eps	min_samples	Outliers	Clusters	% outliers from low-quality sources
0.06	5	194	13	40.7%
0.06	10	635	13	35.1%
0.075	5	66	8	51.5%
0.075	10	197	8	41.6%
0.09	5	18	8	55.6%
0.09	10	51	8	60.8%
0.1	5	11	8	54.5%
0.1	10	26	8	69.2%
0.5	5	0	7	n/a
0.5	10	0	7	n/a

Table 1. DBSCAN sensitivity on UMAP coordinates. Chosen setting: eps=0.075, min_samples=10.

2. Requirement 1: UMAP visualization

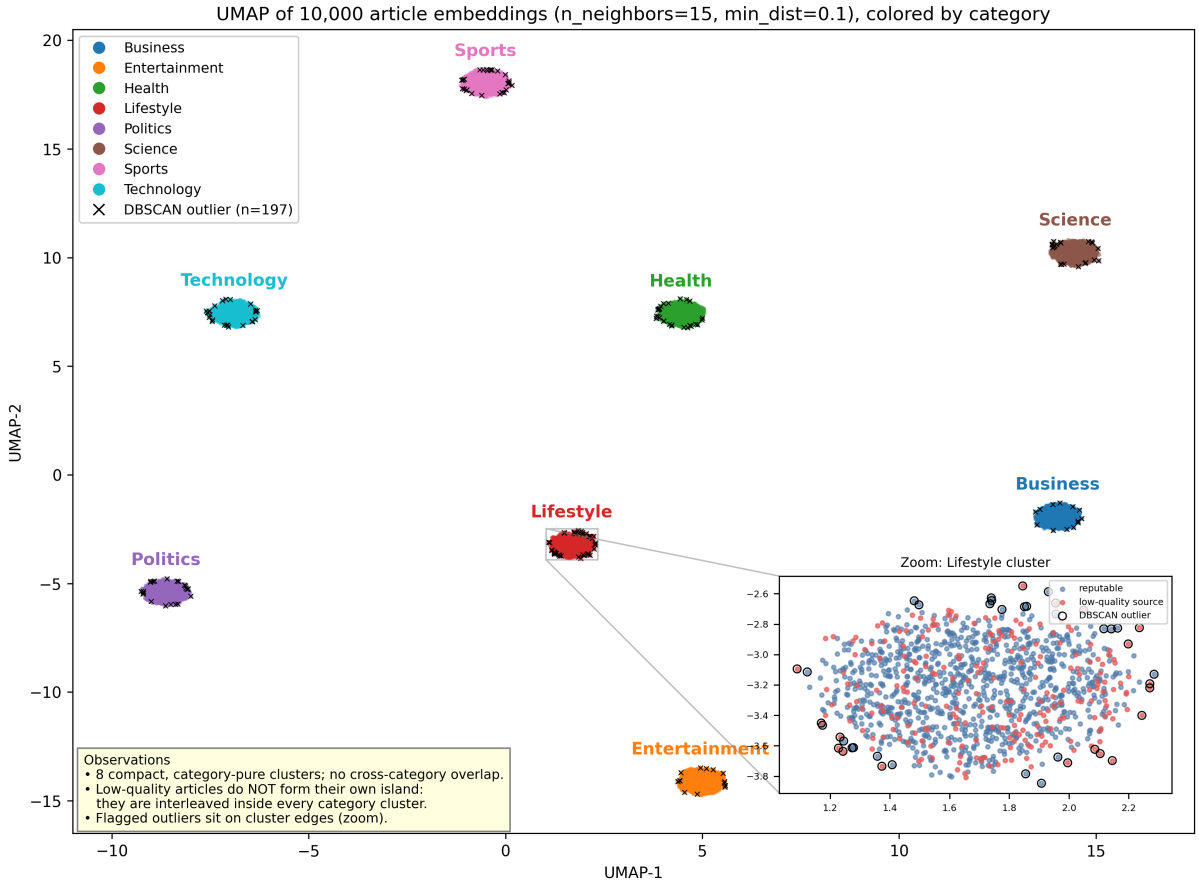


Figure 1. UMAP of all 10,000 articles, color-coded by category. Crosses mark the 197 DBSCAN outliers; the inset zooms on Lifestyle (the cluster with the highest outlier rate), colored by source quality.

Observations on category separation and overlap

- **Separation is excellent.** The 8 categories form 8 distinct islands; silhouette on the 2-D coordinates is 0.96. There is no Sports/Technology or Entertainment/Business overlap in this projection.
- **The problem is not category confusion; it is dispersion.** Red low-quality points are scattered throughout each cluster (inset), not isolated. UMAP preserves local neighborhoods, so a noisy vector still lands next to its nearest same-category neighbors, which hides how far off it really is (see Section 4, H5).
- **Outliers sit on cluster edges.** The crosses ring each blob. Lifestyle shows the most (42 outliers, 3.52%).

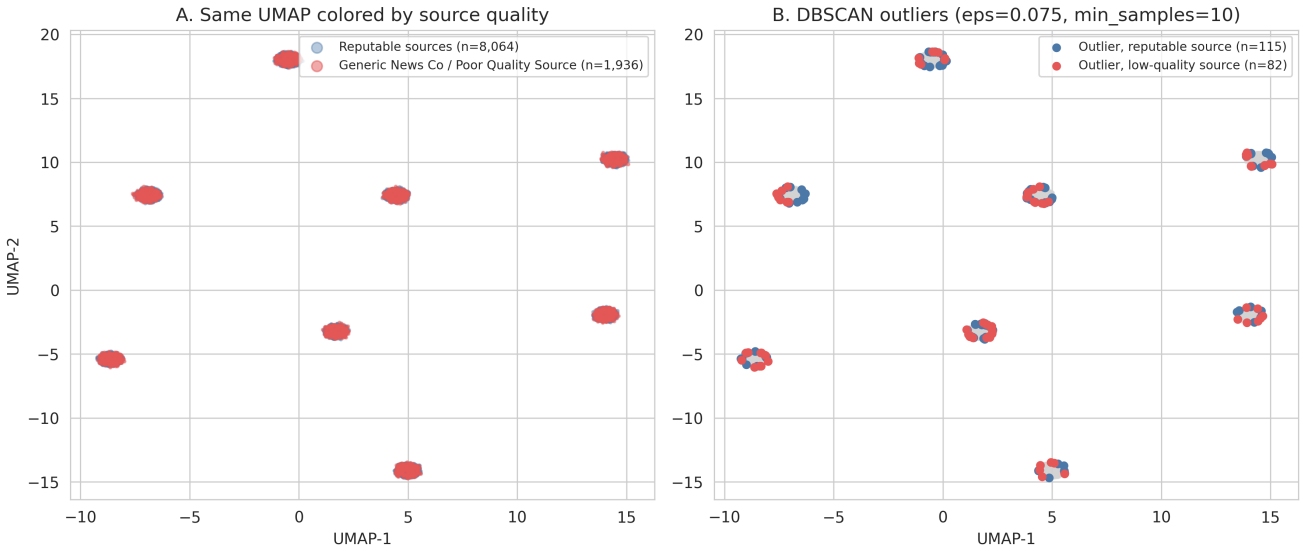


Figure 2. Left: the same embedding colored by source quality. Right: DBSCAN outliers split by source quality; 82 of 197 come from the two low-quality publishers.

3. Requirement 2: Outlier analysis report

Total: 197 outliers = **1.97%** of 10,000 (label = -1), meeting the 50+ target. 8 clusters found.

Outlier distribution

Category	Articles	Outliers	Outlier rate	Low-quality share
Lifestyle	1192	42	3.52%	20.2%
Sports	1280	29	2.27%	18.1%
Health	1270	27	2.13%	20.6%
Technology	1252	25	2.00%	19.4%
Politics	1232	23	1.87%	20.1%
Science	1283	23	1.79%	17.4%
Entertainment	1271	15	1.18%	19.2%
Business	1220	13	1.07%	20.0%

Table 2. By category. Rates differ (chi-square p=0.0007), but the low-quality share is similar across categories (17-21%), so source quality does not explain the category gap.

Source	Articles	Outliers	Outlier rate	Share of all outliers
Generic News Co	954	43	4.51%	21.8%
Poor Quality Source	982	39	3.97%	19.8%
Sports Insider	907	17	1.87%	8.6%
Science Today	923	17	1.84%	8.6%
Business Weekly	858	14	1.63%	7.1%
TechNews Daily	939	13	1.38%	6.6%
Lifestyle Guide	896	12	1.34%	6.1%
Entertainment Tonight	926	12	1.30%	6.1%
Quality Publisher	871	11	1.26%	5.6%
Health Magazine	859	10	1.16%	5.1%
Political Review	885	9	1.02%	4.6%

Table 3. By publisher. The two low-quality publishers have the highest rates (4.51% and 3.97%) but contribute 41.6% of outliers, not the 40%+ "single bad publisher" pattern the brief anticipates; the rest is diffuse edge noise from reputable sources (about 1-2% each).

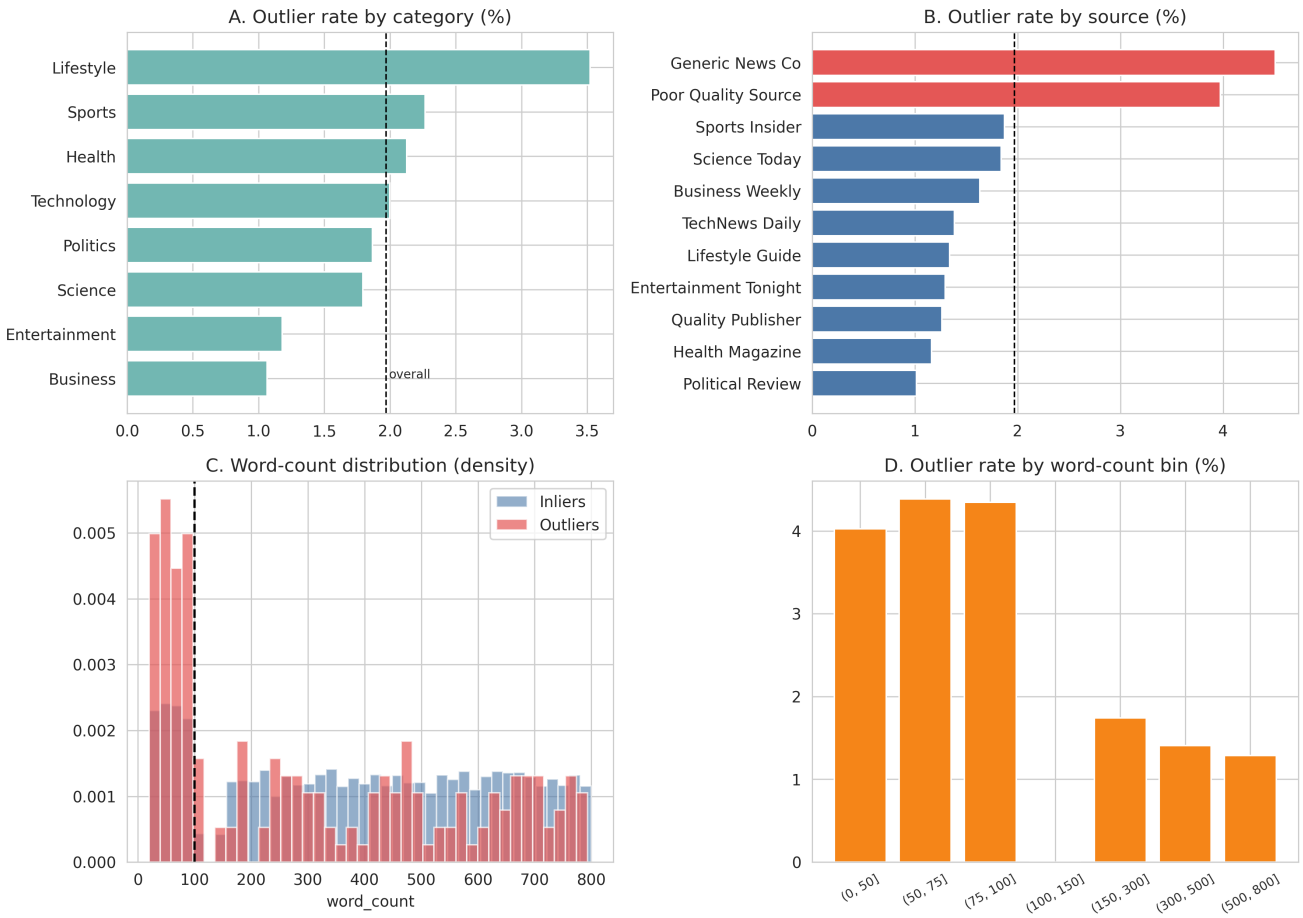


Figure 3. Outlier rate by category and source, word-count distributions, and outlier rate by length bin.

Metadata pattern analysis

- **Missing fields:** 0 null or empty descriptions. But **1,936 descriptions (19.4%) are 4-word stubs**, and the same 1,936 articles have placeholder titles ("Article N", 0 elsewhere). These are effectively missing content that passes a null check.
- **Word count:** 41.6% of outliers have 100 words or fewer, 28.4% fewer than 75, 14.7% fewer than 50. Outlier median length is 241 words vs 400 for inliers. Articles under 75 words are outliers at 4.27% vs 1.62% for the rest (2.6x). Rates are flat (4.0-4.4%) across all three short bins, so it is a threshold effect at about 100 words, not a gradient.
- **Source quality:** every article from Generic News Co / Poor Quality Source has word_count at most 100; every other article has at least 150. Descriptions: mean 101 words among outliers.
- **Title patterns:** no special characters and no non-ASCII/multilingual titles anywhere; the only title anomaly is the placeholder pattern above. Titles for reputable sources are templated ("<Category> Update: Important developments in the field - Article N").
- **Category subtopic diversity:** Entertainment has the *second lowest* outlier rate (1.18%), so the hypothesis that it is legitimately diverse and "scattered" is not supported here. Lifestyle is highest (3.52%) and merits a look.
- **Time:** low-quality share is stable by month (16.7-21.3%, p=0.49), so there is no sign of a recent ingestion spike in this extract.

Deliverables: the full list of the 197 outlier article_ids is in Appendix B; a stratified random sample of 40 outliers with metadata is in Appendix A. Note that UMAP-space DBSCAN catches only a small part of the problem: it flagged 82 of the 1,936 low-quality articles (4.2%). See H5 and the raw-space check below.

4. Requirement 3: Hypothesis documentation

Each hypothesis gives the statement, quantitative evidence, a proposed test, and the link between the metadata pattern and the embedding anomaly.

H1: Two low-quality publishers drive the degradation	**Status: Strongly supported (but confounded with H2/H3)**
Hypothesis	Articles from Generic News Co and Poor Quality Source produce diffuse, off-centroid embeddings and are outliers at about 3x the rate of other sources, because they supply stub content.
Evidence	UMAP-DBSCAN outlier rate 4.24% vs 1.43% (3.0x; p<0.001). 82 of 197 outliers (41.6%) from sources that are 19.4% of the corpus. In 384-d space: mean centroid distance 1.043 vs 0.390 (2.7x); per-article silhouette 0.30 vs 0.75; raw-space DBSCAN flags 1,922 articles, 100% of them from these two sources.
Proposed test	Re-scrape a random 200 articles from each of the two publishers with full text and re-embed. Prediction: centroid distance falls to about 0.39 and they stop being raw-space outliers. If not, the publisher (not the text length) is the cause.
Metadata to embedding link	Publisher feeds supply template stubs, so the encoder receives almost no semantics and returns a vector dominated by noise instead of topic.

H2: Insufficient text (100 words or fewer) causes scattered embeddings	**Status: Supported; cannot be separated from H1 with observational data**
Hypothesis	Articles at or below 100 words, and descriptions of about 4 words, lack the semantic context the encoder needs, so their vectors scatter away from category centroids.
Evidence	1,910 articles under 100 words (plus 26 at exactly 100); 100% of the stub-description articles are at most 100 words. Outlier rate 4.24% (at most 100 words) vs 1.43%. 28.4% of outliers are under 75 words. Rate is flat at 4.0-4.4% across the <50, 50-75, 75-100 bins, and drops to 1.7% above 150 words.
Proposed test	Ablation: truncate 300 reputable articles to 4-word stubs ("Short article about X.") and re-embed. If centroid distance rises from 0.39 toward 1.0, length/content is sufficient to cause the effect without any publisher involvement.
Metadata to embedding link	Fewer tokens means the mean-pooled sentence vector is dominated by generic tokens; the cosine silhouette for these articles is 0.30 vs 0.75.

H3: Placeholder titles remove the last topical signal	**Status: Plausible; fully collinear with H1 so effect size is untested**
Hypothesis	Titles of the form "Article N" carry no category context, so title+description text for these items is nearly information-free.
Evidence	1,936 placeholder titles, all in the two low-quality sources; 0 in other sources. 41.6% of outliers have one. Reputable titles always name the category.
Proposed test	Re-embed low-quality articles with the title removed vs with a synthetic " Update" prefix. If distance to centroid drops sharply with the prefix alone, title quality is a distinct, cheaply fixable cause.
Metadata to embedding link	The encoder sees "Article 17 / Short article about sports." and has only one weak keyword to anchor on.

H4: Category definitions (e.g. Entertainment) are not the root cause	Status: Rejected as primary cause (secondary effect unexplained)
Hypothesis	Category breadth does not explain the anomalies; the modest category differences are geometric, not quality-driven.
Evidence	Outlier rates range 1.07% (Business) to 3.52% (Lifestyle), p=0.0007. Entertainment is 1.18%. Low-quality share is similar in every category (17.4-20.6%). Per-article neighbor category purity is 100%.
Proposed test	Compare Lifestyle vs Business cluster area/density in UMAP and in 384-d (mean within-cluster pairwise distance). Run UMAP with 3 random seeds; if Lifestyle stays highest the effect is real, else it is layout noise.
Metadata to embedding link	Splitting categories would not touch the 19.4% of articles with stub text; the evidence argues against spending effort here first.

H5: UMAP+DBSCAN under-detects the problem because UMAP hides diffuse vectors	Status: Supported
Hypothesis	A neighborhood-preserving 2-D projection pulls noisy vectors toward their nearest category cluster, so 2-D outlier counts understate corpus damage; raw-space detection is the more reliable monitor.
Evidence	UMAP-space DBSCAN flags 82 of 1,936 low-quality articles (4.2% recall). Raw 384-d DBSCAN (eps=0.7, min_samples=5) flags 1,922 (99.3% recall, 100% precision; 14 missed). Silhouette in 2-D = 0.96 vs 0.66 in 384-d (cosine). Re-running UMAP-DBSCAN after cleanup did *not* lower the outlier rate (1.97% to 2.73%), because eps is not comparable when density changes.
Proposed test	Compute trustworthiness / continuity of the UMAP embedding at n_neighbors 5, 15, 50, and compare how many low-quality articles DBSCAN flags at each. Prediction: recall stays low at all settings.
Metadata to embedding link	Anomalous vectors are far from every cluster in 384-d, but UMAP maps them next to whichever cluster they are least far from.

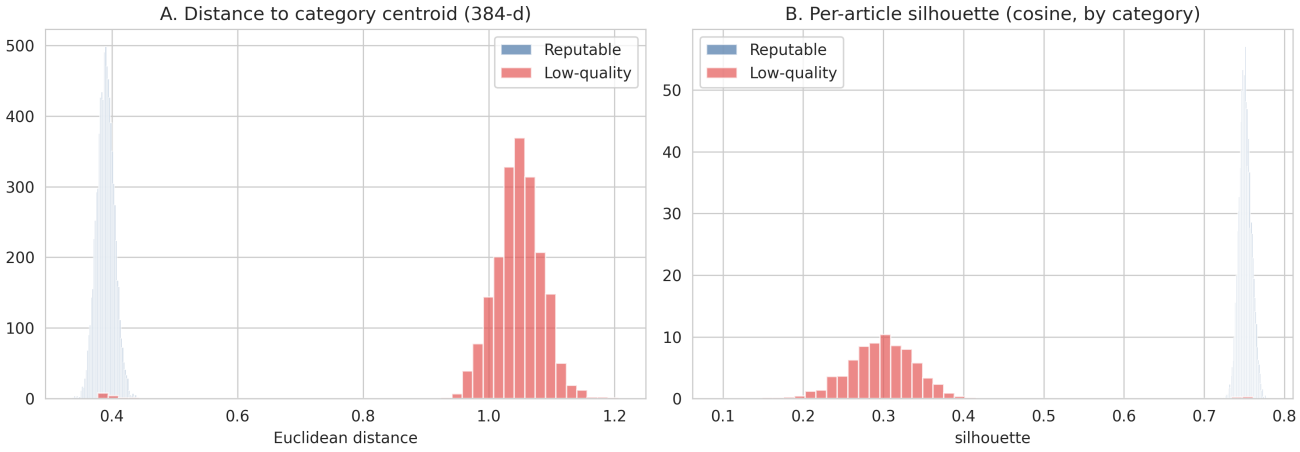


Figure 4. In the original embedding space, low-quality articles (red) are far from category centroids (left) and have much lower silhouette (right).

5. Requirement 4: Cleanup plan and validation strategy

Prioritized cleanup actions

#	Action	Estimated impact	Effort	Why this rank
1	Quarantine Generic News Co and Poor Quality Source from the embedding index; re-scrape full text or drop	Removes 1,936 articles (19.4%) = 100% of the severe embedding degradation; raw-space outliers 1,922 to 0	Low	Largest effect for the least work; reversible
2	Ingestion gates: word_count above 100 (reputable minimum observed is 150), placeholder-title regex (^Article \d+$), description of at least 20 words, plus an embedding gate (distance to nearest category centroid above 0.5, versus 0.44 observed maximum for clean articles) that holds items for review	Prevents recurrence; the metadata gates alone would have caught 100% of the 1,936 stub articles	Low-Med	Stops the problem returning; cheap
3	Backfill titles and descriptions for the quarantined articles from the article body (first 200 characters or an LLM summary), then re-embed	Recovers content instead of losing 19% of the catalog; expected to move them to the clean cluster if H2/H3 hold	Medium	Needs body text; depends on test H1/H2
4	Weekly raw-space monitoring: DBSCAN (eps=0.7) outlier rate and mean centroid distance by source, with alerts	Early warning for the next bad feed	Low	Cheap; avoids reliance on 2-D UMAP counts
5	Investigate Lifestyle (3.52% outlier rate); **do not** split Entertainment	Likely small	Low	H4: evidence does not support category splitting

Table 4. Cleanup actions ranked by estimated impact (impact figures come from the analysis above or the simulated cleanup).

Success metrics and target thresholds

Metric	Baseline (current)	After simulated cleanup	Target
Raw 384-d DBSCAN outlier rate (eps=0.7, min_samples=5)	19.2% (1,922)	0.0% (0)	0.5% or lower
Category silhouette, cosine, 384-d	0.664	0.844	0.80 or higher
Mean distance to category centroid	0.516	0.390	0.45 or lower
Share of catalog from stub-content sources	19.4%	0.0%	under 2%
UMAP-space DBSCAN outlier rate (eps=0.075, ms=10)	1.97%	2.73%*	Report only; re-tune eps each run
Recall@10 (offline)	71%	not measured	Hold at 71% or higher (guardrail)
User engagement	down 18% this month	not measured	Recover at least half (9 points) within 4 weeks

Table 5. *Not a regression: with 19% fewer points, UMAP renders each cluster less dense, so the same eps flags more edge points. This is why UMAP-space outlier rate is not used as a gate. The brief's example silhouette target (0.4 to 0.65) does not apply because the measured baseline here is already 0.66; the sharper signal is in the outlier and centroid-distance metrics.

Validation methodology

- **Re-run UMAP:** after cleanup, same parameters and seed (done in simulation; Figure 5). Clusters stay separate and tidy, but visually the 2-D map barely changes, which is exactly why 2-D alone is insufficient.
- **Cluster purity:** measured at 100% (mean and minimum) for all DBSCAN clusters before cleanup, so purity does *not* discriminate here. Use it as a regression guardrail on live data, where mixed clusters are plausible.

- **Hold-out re-embedding:** run the H1-H3 tests on 200-300 article samples before changing production, so the fix is credited to the right cause.
- **Neighbor audit:** for 500 random seed articles, human-judge top-10 recommendations before and after; the offline recall@10 metric alone already proved insufficient.

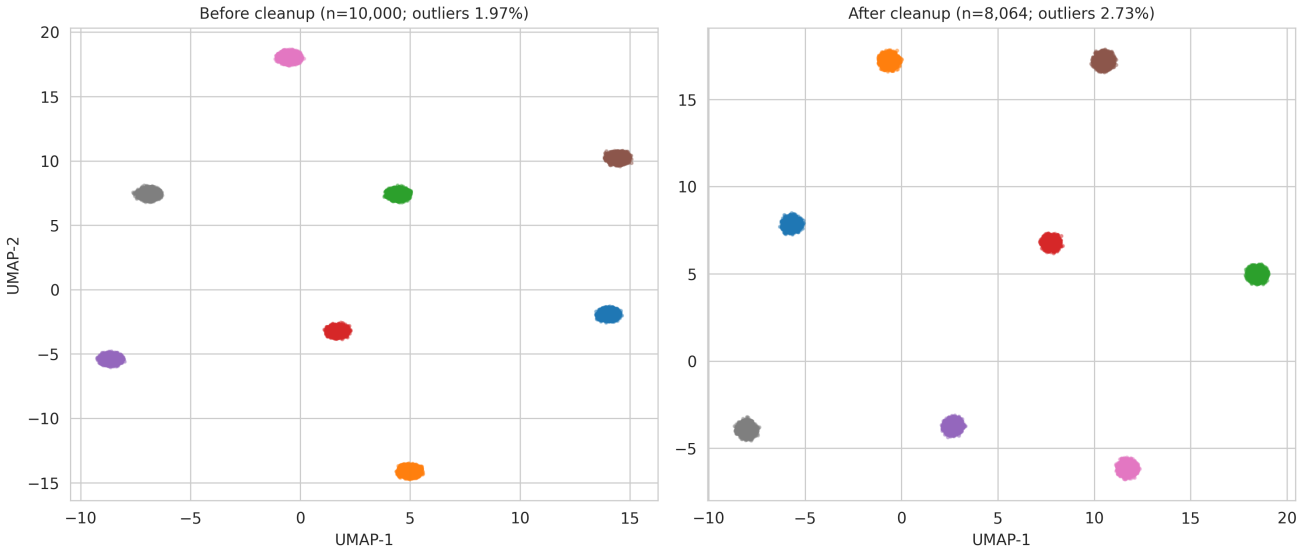


Figure 5. UMAP before and after removing the 1,936 stub-content articles (same parameters, seed 42).

A/B testing recommendations

- **Design:** 50/50 user-level randomization. Control = current index. Treatment = cleaned index (quarantine + gates). Run 2 weeks minimum, covering a full weekly cycle.
- **Primary metric:** session engagement (the metric that dropped 18%). **Secondary:** click-through on recommendations, share of sessions with a "bad rec" report, category-mismatch rate of served items.
- **Guardrails:** catalog coverage (removing 19.4% of articles shrinks supply; confirm Business/Lifestyle feeds do not run thin), latency, publisher-level exposure.
- **Staged rollout:** 5% then 25% then 50%, with the raw-space outlier rate and centroid-distance monitors as automatic stop conditions.
- **Sequence:** run the H1-H3 ablations in the first 72 hours; if H1 holds, ship action 1 immediately because it is reversible and low-cost.

6. Conclusion and limitations

The evidence points to data quality, not embedding model failure: 19.4% of the catalog arrives with stub titles and 4-word descriptions from two publishers, and those vectors are about 2.7x farther from their category centroid than clean ones. Fixing ingestion is cheap and measurable, so the recommendation to the board is to invest in data infrastructure.

Limitations to keep in view

- **Confounding:** publisher, length, and placeholder titles are perfectly collinear in this extract, so H1-H3 are all supported but not individually isolated until the ablations are run.
- **Tuning:** the UMAP-DBSCAN outlier count depends heavily on eps and min_samples; 197 is one defensible setting, not a ground truth.

- **Synthetic data:** templated text and tight clusters mean cluster purity (100%) and cross-category purity (100%) are unrealistically clean; real data will almost certainly show category mixing that this extract does not.
- **User-visible link not shown:** I did not reproduce the "sports in tech feeds" symptom from nearest-neighbor lists. The recommender (ranking, ANN index, user-profile averaging) must be inspected to confirm how degraded vectors reach users.
- **Simulation only:** the after-cleanup numbers come from deleting the stub articles, not from re-embedding repaired text; true gains from backfilling depend on H2/H3.

Appendix A. Random sample of 40 outliers with metadata

article_id	Category	Source	Words	Desc. words	Dist. to centroid	Silhouette
ART01150	Sports	Business Weekly	328	114	0.38	0.76
ART02158	Science	Generic News Co	98	4	1.04	0.28
ART02202	Sports	Generic News Co	99	4	1.06	0.27
ART02475	Health	Poor Quality Source	98	4	1.09	0.28
ART02490	Health	Poor Quality Source	52	4	1.07	0.29
ART03280	Technology	Generic News Co	94	4	1.06	0.31
ART03749	Politics	TechNews Daily	493	171	0.40	0.74
ART04930	Lifestyle	Generic News Co	99	4	1.03	0.27
ART05435	Lifestyle	TechNews Daily	685	247	0.39	0.75
ART05548	Politics	Generic News Co	58	4	1.02	0.34
ART05707	Science	TechNews Daily	263	95	0.38	0.76
ART05976	Sports	Science Today	475	171	0.38	0.76
ART06001	Technology	Generic News Co	48	4	1.02	0.35
ART06055	Technology	TechNews Daily	705	266	0.40	0.75
ART06144	Politics	Sports Insider	424	152	0.39	0.75
ART06193	Sports	Quality Publisher	531	190	0.41	0.75
ART06200	Business	Generic News Co	49	4	1.04	0.30
ART06464	Sports	Sports Insider	692	247	0.39	0.76
ART06523	Business	Poor Quality Source	63	4	1.02	0.35
ART06715	Science	Business Weekly	370	133	0.38	0.77
ART06838	Sports	Sports Insider	663	247	0.38	0.76
ART07401	Sports	Generic News Co	25	4	1.01	0.35
ART07497	Sports	Entertainment Tonight	569	209	0.39	0.76
ART07506	Technology	Business Weekly	793	285	0.39	0.75
ART07539	Health	Generic News Co	37	4	0.99	0.37
ART07696	Lifestyle	Poor Quality Source	60	4	1.10	0.27
ART07779	Health	Generic News Co	59	4	1.07	0.28
ART08085	Science	Science Today	681	247	0.40	0.76
ART08335	Lifestyle	Entertainment Tonight	341	114	0.40	0.75
ART08378	Technology	Health Magazine	772	285	0.40	0.75
ART08616	Technology	Generic News Co	58	4	1.04	0.31
ART08671	Lifestyle	Poor Quality Source	37	4	1.06	0.28
ART08886	Health	Poor Quality Source	22	4	1.05	0.31
ART08963	Technology	Science Today	565	209	0.40	0.75
ART09168	Health	Political Review	759	285	0.39	0.75
ART09345	Entertainment	Lifestyle Guide	760	285	0.38	0.76
ART09380	Sports	Science Today	226	76	0.38	0.76
ART09614	Science	Generic News Co	79	4	1.04	0.33
ART09628	Science	Science Today	544	190	0.39	0.77
ART09869	Health	Entertainment Tonight	414	152	0.40	0.74

In this sample, 18 of 40 are from low-quality/short articles; the other 22 are reputable-source articles flagged on cluster edges (centroid distance about 0.39, silhouette about 0.75). These are false positives of edge-based detection, not data defects.

Appendix B. Exported outlier article IDs (n=197)

ART00153	ART03734	ART05578	ART06535	ART07545	ART08583	ART09345
ART00465	ART03749	ART05604	ART06541	ART07555	ART08616	ART09380
ART00628	ART03820	ART05707	ART06619	ART07571	ART08671	ART09413
ART00634	ART04075	ART05749	ART06624	ART07696	ART08682	ART09424
ART00649	ART04090	ART05762	ART06627	ART07699	ART08729	ART09465
ART00886	ART04098	ART05785	ART06632	ART07728	ART08751	ART09501
ART00948	ART04100	ART05803	ART06715	ART07779	ART08770	ART09503
ART01125	ART04228	ART05835	ART06820	ART07792	ART08780	ART09550
ART01144	ART04279	ART05870	ART06838	ART07887	ART08786	ART09567
ART01150	ART04336	ART05976	ART06874	ART07895	ART08815	ART09607
ART01427	ART04345	ART06001	ART06875	ART07902	ART08886	ART09614
ART01430	ART04419	ART06055	ART06889	ART07925	ART08894	ART09624
ART01628	ART04744	ART06081	ART06934	ART08014	ART08897	ART09628
ART01997	ART04768	ART06111	ART06943	ART08026	ART08904	ART09671
ART02035	ART04795	ART06115	ART06976	ART08047	ART08905	ART09850
ART02158	ART04923	ART06144	ART07010	ART08084	ART08908	ART09857
ART02202	ART04930	ART06182	ART07020	ART08085	ART08947	ART09869
ART02402	ART05001	ART06193	ART07112	ART08091	ART08963	ART09905
ART02475	ART05048	ART06200	ART07155	ART08100	ART09011	ART09916
ART02490	ART05097	ART06219	ART07193	ART08189	ART09024	ART09932
ART02778	ART05103	ART06248	ART07217	ART08308	ART09042	ART09984
ART02807	ART05171	ART06350	ART07259	ART08309	ART09045	ART09985
ART02863	ART05209	ART06421	ART07278	ART08311	ART09074	ART09991
ART03201	ART05233	ART06444	ART07377	ART08335	ART09109	
ART03280	ART05325	ART06464	ART07396	ART08378	ART09166	
ART03297	ART05364	ART06478	ART07401	ART08390	ART09168	
ART03366	ART05435	ART06513	ART07497	ART08410	ART09216	
ART03649	ART05548	ART06523	ART07506	ART08484	ART09271	
ART03680	ART05570	ART06533	ART07539	ART08549	ART09294	